

# SAITO ADMIN

## Full Project Briefing for the Incoming Manager

Complete system map: what Saito is, how it is built, how it is governed, where the project stands right now, and the decisions that are on the table.

|                        |                                                                         |
|------------------------|-------------------------------------------------------------------------|
| Date                   | 2026-09-18 (P-9 freeze, Baku UTC+4); briefing refreshed 2026-09-19      |
| Prepared by            | Accio (project AI agent, phases P-0 through P-9)                        |
| Classification         | Internal - live system state; no credentials included                   |
| Codebase state         | git main: P-9 FROZEN (freeze commit on top of 6d7a883e, the P-8 freeze) |
| Live system            | Supabase Postgres (pooler :6543) + Next.js dev server :3000             |
| Companion docs in repo | HANDOVER.md, ARCHITECTURE.md, P9_*.md, MASTER_FEATURE_MAP.md            |

Every number in this document was verified read-only on 2026-09-19 against the live database, the git repository, and the gate execution logs (the full reflow ran 2026-09-18). The F/O contract-conflict decision (Section 7) was ratified and resolved the same day; P-9 is FROZEN. Open items: the P-10 terminal-provider decision (Q7) and the Wave-A feature plan (MASTER\_FEATURE\_MAP.md).

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# 1. What Saito Admin is

Saito Admin is a **complete restaurant POS and back-office suite**: floor plans and table management, order lifecycle with split/partial payments and a strict payment state machine, kitchen display system (KDS), reservations with pre-orders, cash drawer with an audited ledger, staff economy (shifts, time clock, payroll, tips, schedule), inventory with recipes and stock deduction, multi-location foundation, and an analytics/AI layer. It is a production system with real operational history - not a demo.

## 1.1 Technology and deployment

| Layer          | What                                                      | Where                                                           |
|----------------|-----------------------------------------------------------|-----------------------------------------------------------------|
| Frontend + API | Next.js 16 (App Router), TypeScript                       | artifacts/saito-admin/ (dev: localhost:3000)                    |
| Database       | PostgreSQL on Supabase (project jbxmInsicbfkbsatnoej)     | psql via pooler :6543; PostgREST /rest/v1                       |
| Business logic | PostgreSQL: SECURITY DEFINER atomic RPCs + triggers + RLS | supabase/migrations/ (60+ applied)                              |
| Access         | Staff PIN login; session cookie + double-submit CSRF      | src/lib/api-auth.ts, csrf.ts, shiftLock.ts, location-context.ts |
| State sync     | Supabase Realtime + outbox_events delivery                | DB is authoritative; client = presentation + resync             |

## 1.2 Scale (measured 2026-09-17/18)

| Metric                         | Value                                                                                | Source                               |
|--------------------------------|--------------------------------------------------------------------------------------|--------------------------------------|
| API route files                | 180 (0 TODO/FIXME/stub, grep-verified)                                               | BACKEND_COMPLETION_MAP_2026-09-17.md |
| Public DB tables               | 138 (post-P-9: dead tables archived as 17 p9_archive_%, net of M2-M5)                | pg_class scan, live 2026-09-19       |
| Orders / order items           | 729 / 701                                                                            | live counts 2026-09-19               |
| Outbox events / operation logs | 6,355 / 1,862                                                                        | live counts 2026-09-19               |
| Real staff (DENY53) / active   | 53 real (9 active, 9 pinned; login pool = 9); 133 total incl. 80 inert gate fixtures | P-9 M7 + S5.4a-d; live 2026-09-19    |
| Frozen gates                   | 15 (A, E/S, F, O, K-L3, K-L4, P-1..P-8, P-9)                                         | repo root .cjs files                 |
| P-9 migrations applied         | 8 (M1a, M2-M7, M1b) incl. 10 new FKs + 17 p9_archive_% tables                        | supabase/migrations/2026091800000*   |

## 2. Architecture: the rules the whole system obeys

*These rules are ratified and frozen (ARCHITECTURE.md + HANDOVER section 4). Breaking any of them is a scope violation that requires a new freeze cycle. This is the single most important thing for a manager making code-level decisions to internalize.*

### 2.1 Single Source of Truth (SSOT)

- Every business entity has exactly ONE authoritative table (orders, order\_payments, order\_items, inventory\_transactions, reservation\_\*, table\_floors state, operation\_logs). No denormalized copies: orders.items JSON is deprecated; table\_floors snapshot columns are convenience only.
- All business logic lives in the database. Every multi-table business operation is one **SECURITY DEFINER atomic RPC**: lock affected rows FOR UPDATE, validate preconditions, mutate in a single transaction, write an audit row, return JSONB. Next.js API routes are **thin proxies**: auth, CSRF, shift check, then one RPC call. Never business logic, never multi-step mutations.

### 2.2 The lifecycles derive and synchronize - they never merge

```
TABLE:      empty -> occupied -> (paid) -> dirty -> empty      (dirty = post-payment cleanup, BY DESIGN)
ORDER:      new/open/confirmed -> in_kitchen -> ... -> paid -> closed
            (+ cancelled / refunded / partially_refunded / voided; reopen = authorized full reversal)
KITCHEN:    pending -> accepted -> preparing(cooking) -> ready -> served    (order_items.kitchen_status)
PAYMENT:    AUTHZ(P-1) . STATE(P-2) . IMMUTABLE(P-3) . IDEMPOTENT(P-4) . ATOMIC(P-5)
            . REFUND/VOID/REOPEN(P-6) . CONCURRENT(P-7) . CASH-DRAWER(P-8)    (each layer frozen + gated)
DERIVATION: full payment writes table dirty + clears pointer; last-open-order cancellation frees the table
            kitchen chips go to table.kitchen_status (own trigger) - NEVER to table.status
            new seat on a dirty table is server-blocked until Clear
```

### 2.3 Security model (frozen)

- **Identity**: staff PIN login (PBKDF2, server-side). The saito\_token cookie carries staff + role + organization + location; the server ALWAYS re-derives identity and location from the session (resolveWriteLocationContext) - client-supplied location\_id is never trusted (P-1).
- **Authorization**: has\_permission(uuid, text) is the permission SSOT (P-1); requirePermission on routes; manager-override gates (refund.approve / void.approve) for money reversals; requireActiveShift for floor operations; fail-closed (NULL location = deny).
- **Integrity**: DB triggers enforce state machines and immutability (payment state guard, payment immutability, underpayment block, paid-requires-record, table pointer validation, table archive guard). RLS on tables; the service\_role key is only ever on the server. CSRF: double-submit (x-csrf-token header must equal the saito\_csrf cookie).
- **Audit and sync**: every money path writes canonical audit + outbox\_events (scoped delivery); realtime channels only trigger client resync. The DB is authoritative.

### 2.4 The "F-contract" (test discipline that shaped everything)

Established in the F module and still binding: test fixtures are **never deleted from staff** - they are neutralized (is\_active=false, pin cleared) and left inert; test tables/orders are removed; every gate is idempotent and zero-residue; a crashed gate must leave a state its own START cleanup can recover. This contract is exactly what created the current decision case (Section 7): frozen gates written before P-9 added a new foreign key now disagree about teardown ORDER, not about behavior.

### 3. Governance: the freeze-and-audit workflow

This is how the project actually moves. A manager who ignores this workflow will break things; one who uses it can change anything - at a known cost.

#### 3.1 The core loop (one phase = one cycle)

|                   |                                                                                        |
|-------------------|----------------------------------------------------------------------------------------|
| 1 READ-ONLY AUDIT | map the live surface (tables, RPCs, triggers, RLS, routes) with evidence               |
| 2 DECISION MATRIX | findings F-1..F-n, each with options + recommendation; NOTHING applied yet             |
| 3 RATIFICATION    | the owner ratifies decisions one-by-one ("D-1 GO") - never batched silently            |
| 4 MIGRATION(S)    | single transaction, ON_ERROR_STOP=1, preflight (orphans = 0), ROLLBACK written FIRST   |
| 5 GATE            | a live test battery (.x-gate.cjs) proves the new contract against REAL DB + dev server |
| 6 FULL REFLOW     | every previously frozen gate re-run (never assumed green); identical tallies required  |
| 7 FREEZE + COMMIT | the gate becomes frozen; git commit + push; HANDOVER updated                           |

#### 3.2 What "frozen" means (binding on all future work)

- A frozen gate file (.cjs) and its checked behavior may not change without an explicitly ratified **correction cycle**: proposed diff as a file-level document + source-level proof (trigger bodies, FK catalog rows, past green runs) + a separate GO. The 14 frozen gates are the regression net for the entire system.
- Frozen DB contracts (triggers, RPC signatures, state registries, RLS, the 10 P-9 FKs) may not be dropped or weakened (e.g. RESTRICT to CASCADE) without ratification. "Cascade is easier" is a rejection reason on its own - it changes a ratified safety decision.
- DO-NOT-REOPEN list (HANDOVER section 2): authz/token-first fns, server-derived location, KDS finalized-order guards, stock consumption in mark\_order\_ready (load-bearing), table-lifecycle separation (no occupied-to-in\_kitchen edges), shift no-overlap invariant. Plus 4 documented frozen-boundary residuals (Section 4.5) that must not be "silently fixed".

#### 3.3 Evidence rules (why every claim carries a number)

- Every quantitative claim is traceable to a read-only query, a gate log line, or a file diff. Gates emit PASS / **REAL-RISK** (product invariant failed) / **HARNESS-FAILURE** (test machinery failed - fix the harness, never the product) / PASS-with-expectations (known, documented). Zero-residue is asserted by the gate itself at the end (T.1-style checks).
- Any deviation from expected tallies = **STOP + triage**: the new work is the suspect until proven environment/harness. The owner then decides: in-protocol fix (with evidence, logged) or hard stop with options. There is no silent override of a frozen contract - not even a "small" one-line change.
- Corrections to frozen gates are separate ratified cycles - exactly what happened to gate F that morning (resolved with a 2-line swap, now 35/35) and what then happened to gate O the same afternoon (O-A + O-B ratified, applied, 38/38 - the full record is in Section 7).

## 4. Phase history: P-0 through P-8 (all frozen)

The project hardened in phases. Each phase ran the governance loop: read-only audit, ratified decisions, migrations, gate, full reflow, freeze. All of it is committed at [git 6d7a883e](#) (branch main, main == origin). The module gates A/E/S/F/O/K predate the P-series; the P-series froze the payment/finance boundary end to end.

### 4.1 The 14 frozen gates (baseline tallies - "the net")

| Gate file         | Module / contract                                           | Baseline             | Re-verify command (repo root)                 |
|-------------------|-------------------------------------------------------------|----------------------|-----------------------------------------------|
| .a-regression.cjs | A - auth, sessions, PIN login, logout, overrides            | 39/39                | node .a-regression.cjs                        |
| .es-gate.cjs      | E/S - staff, roles, time clock, shifts, cash-rpc signatures | 54/54                | node .es-gate.cjs                             |
| .f-gate.cjs       | F - floors/tables registry, lifecycle guards, dirty state   | 35/35                | node .f-gate.cjs                              |
| .o-gate.cjs       | O - order state machine, tables, payments-create, QR        | 38/38                | O_ROUTE=/api/orders/route.ts node .o-gate.cjs |
| .k-l3-probe.cjs   | K - KDS item lifecycle L3                                   | 8/8                  | node .k-l3-probe.cjs                          |
| .k-l4-probe.cjs   | K - KDS L4 (finalized-order guard, stock)                   | 16/16                | node .k-l4-probe.cjs                          |
| .p1-gate.cjs      | P-1 - authorization: permission SSOT + location scope       | 29/29                | node .p1-gate.cjs                             |
| .p2-gate.cjs      | P-2 - 12-state payment machine registry + guard             | 13/13                | node .p2-gate.cjs                             |
| .p3-gate.cjs      | P-3 - amount immutability, underpayment block, reopen flag  | 25/25                | node .p3-gate.cjs                             |
| .p4-gate.cjs      | P-4 - idempotency (exactly-once pay/refund, 409/400)        | 19/19                | node .p4-gate.cjs                             |
| .p5-gate.cjs      | P-5 - atomicity: paid requires a payment record             | 10/10                | node .p5-gate.cjs                             |
| .p6-gate.cjs      | P-6 - refund / void / reopen (authorized full reversal)     | 15/15                | node .p6-gate.cjs                             |
| .p7-gate.cjs      | P-7 - concurrency matrix (pay/pay, pay/cancel, ...)         | 16/16                | P7_ALL=1 node .p7-gate.cjs                    |
| .p8-gate.cjs      | P-8 - cash drawer ledger SSOT, close v2, clock, breaks      | 8/8 (2P+6EC, 0 risk) | P8_ALL=1 node .p8-gate.cjs                    |

### 4.2 What each P-phase froze (the substance)

- **P-1 Authorization** (2026-09-12): has\_permission() = permission SSOT; fail-closed location assertion on the financial write path (complete\_payment\_atomic\_v2 gained p\_location\_id; order location must equal session location; NULL = deny); 4 broken RPCs retired; owner role 65/65 permissions; /api/orders/void gated on pos.void. Killed the class of bug where a cashier could pay or refund at a location they are not assigned to.
- **P-2 State** (2026-09-12): the 12-state payment machine (22 transition rules) registered in state\_transitions as the single SSOT (it was hardcoded in one function); the stale 8-state check dropped (it pre-blocked 4 legal states); guard trigger on order\_payments.status. No state outside the registry can be written.
- **P-3 Amounts** (2026-09-13): payment records immutable (UPDATE of amount/method/is\_refund blocked; DELETE blocked except the trusted reopen path under the transaction-scoped app.payment\_ledger\_reopen flag); an order cannot enter paid/closed while 0 < paid < total (underpayment block). Legacy reconciliation gaps (342 paid-without-record + 42 mismatch, all pre-2026-09-09) frozen, no backfill - ratified.
- **P-4 Idempotency** (2026-09-13): payment\_idempotency\_keys with (namespace, key) PK; same-key different-payload = 409 with zero mutation; keyless pay/refund = 400; 30-day TTL + pruner. Root-caused and fixed a PostgREST/pgbouncer gateway hang (error code 40001 made the gateway hang; the proven 409 path uses P0001).
- **P-5 Atomicity** (2026-09-13): "no paid-without-record" enforced twice (the order-transition RPC rejects direct paid/refunded transitions AND a DB trigger asserts a captured payment row exists on to-paid); payment records cannot be re-attached to a different order. Legacy drift frozen at 274/54/13/4 with a zero-NEW-drift assertion in the gate. (The legacy-drift data decision is P-9 adjacent - the forward contract is airtight.)

- **P-6 Refund/Void/Reopen** (2026-09-13): reopen = FULL financial reversal, authorized only with token-derived identity (caller identity cross-checked) + orders.edit + location scope + MANAGER OVERRIDE; the legacy 4-arg overload is frozen to postgres-only; over-refund proven impossible under concurrency (net-paid cap under FOR UPDATE); audit + outbox now emitted on every reversal (previously money could move silently).
- **P-7 Concurrency** (2026-09-15..17): dependency-contamination gate (which writers can reach the money tables; 4 NO / 2 YES ratified) then a live 16/16 concurrency battery: pay/pay, pay/cancel, pay/new-seat-on-dirty-table, pay/dismiss, drawer-close vs payment, reopen vs pay - exactly one wins, zero mutation on the loser.
- **P-8 Cash drawer** (2026-09-17): cash drawer ledger SSOT (cash\_drawer\_sessions + cash\_drawer\_log, close-row = 1 per session), close\_cash\_register\_v2 (idempotent replay, manager\_pin D-5, outbox D-12, FOR UPDATE D-3), time clock + breaks + auto-clockout. Frozen at 6d7a883e with the full reflow green.

### 4.3 The pre-P module gates (A / E/S / F / O / K)

A: auth/sessions/lockout/self-approve guard. E/S: staff economy incl. the cash-RPC signature contract (54/54). F: the procedural floor/table lifecycle (empty/occupied/reserved/dirty, archive guard trigger, seat/transfer/clear atomics). O: order state machine + table binding + payments-create + QR insert path. K: KDS (kitchen tickets, item lifecycle, finalized-order guard; stock consumption in mark\_order\_ready is PRESERVED and load-bearing - do not "modernize" it).

### 4.4 Build/quality expectations (frozen)

- npx next build: exit 0 (270 static pages). npx tsc --noEmit: 0 errors (non-test). The dev server must be running (npm run dev in artifacts/saito-admin) for any gate: gates hit the live DB AND localhost:3000.

### 4.5 Known frozen-boundary residuals (documented, deferred - do NOT silently fix)

- F-03: 3 fns filter by table\_number without location\_id - safe today because 0 table\_numbers exist in two locations (single-location deployment).
- "dirty" is intentionally OFF the state\_transitions registry (operational post-payment state; enforced by the seat guard + clear\_table\_atomic).
- order\_items.kitchen\_status holds 89 hot + 9 bar + 2 sushi station values (pre-existing data hygiene; station is a separate column).
- idx\_orders\_active\_table UNIQUE(table\_number) is not location-scoped (same single-location rationale).

### 4.6 Known pre-existing issues (flagged in gate reports; owned by future security passes)

- payment\_idempotency\_keys is world-readable (anon/authenticated SELECT incl. financial result jsonb) - flagged in P-4.
- order\_payments legacy ACL grants (anon rwd) - functional contract unaffected; flagged in P-4.
- The refund route maps REFUND\_EXCEEDS\_PAID to HTTP 500 instead of 4xx - the invariant holds; the mapping is a cleanup candidate (flagged in P-6).

## 5. P-9: schema normalization (FROZEN 2026-09-18)

P-9 is COMPLETE and FROZEN (2026-09-18). Mandate (ratified 2026-09-17, see P9\_IMPLEMENTATION\_PLAN\_2026-09-17.md): normalize the schema debt that accumulated under the frozen contracts - duplicated settings writes, legacy/frozen tables, missing foreign keys, and a staff table holding 1,108 rows of which only 53 were real. **P-9 adds contracts; it does not change frozen behavior.** Every step ran the governance loop: audit (P9\_SCHEMA\_AUDIT\_2026-09-17.md) - decision matrix - ratification - migration - gate.

### 5.1 Applied migrations (single-transaction, rollbacks written first, live-verified)

| File         | What                                                                                                                                                                                                        | Key numbers                                                |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| M1a (000001) | RLS/GRANT normalization pass: settings contract - direct authenticated access revoked on the 6 v_settings_* write surfaces (incl. reservation_preorder_items); settings writes go through the contract path | grant matrix verified both ways (S1.1-S1.6)                |
| M2 (000002)  | Settings views: decomposition of duplicated settings writes (10 dups removed)                                                                                                                               | 10 v_settings_* views = settings (S2.1-S2.3)               |
| M3 (000003)  | app_settings dead columns/rows cleanup                                                                                                                                                                      | S3 post-drop smoke green                                   |
| M4 (000004)  | Legacy duplicate column drop (orders_items)                                                                                                                                                                 | S3 green                                                   |
| M5 (000005)  | Dead tables archived: CREATE TABLE p9_archive_ AS SELECT * then DROP (audit_log mirror trigger removed first; drop set corrected via M5-R1/R2)                                                              | 17 p9_archive_% tables live; no consumer broken            |
| M6 (000006)  | 10 safe FKs added (7 CASCADE / 6 SET-NULL class / 1 RESTRICT) - each with orphan = 0 proven before apply; tripwire-checked (S4.1); set corrected 14 to 10 via M6-R1                                         | the RESTRICT one: table _floors.current_order_id -> orders |
| M7 (000007)  | Staff fixture purge: 1,108 to 53 real (neutralize, never delete); 13-step RESTRICT-aware child delete; all 1,055 fixture IDs archived; dining_groups + merge_tables_v4 intact                               | pool 39 to 9; 565 child rows; 0 orphans; preflight 24/24   |
| M1b (000008) | M1 remainder: residual public/anon auth-surface revokes                                                                                                                                                     | S1.1-S1.6 green                                            |

### 5.2 The P-9 gate (.p9-gate.cjs - NEW, 25 checks - FINAL 25/25, 0 REAL-RISK, 0 HARNESS)

| Section   | What it proves live                                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------|
| S1.1-S1.6 | RLS/GRANT matrix (M1 + M1-R1) both directions                                                                                        |
| S2.1-S2.3 | 10 v_settings_* views equal settings; 8 settings routes 200; geo route 405 (frozen shape)                                            |
| S3.1-S3.4 | Post-drop smoke: no consumer of removed surfaces breaks                                                                              |
| S4.1      | FK tripwire: a violating write is rejected (the M6 FKs are live, not decorative)                                                     |
| S5.1-S5.5 | Purge proof: fixture patterns inert; active = 9; pool = exactly 9; 53-ID denylist intact; fresh-fixture login works and is torn down |
| S6.1      | test_rls_role contract intact                                                                                                        |
| S7.1      | P-8 cash-drawer canary (T-4 shape: close then idempotent true then new key rejected; ledger close-row = 1)                           |
| T.1       | Zero residue: 0 P9GATE_* staff after the run                                                                                         |

### 5.3 What P-9 fixed along the way (harness lessons now baked into the gate)

- Supabase pooler role anomaly (transient authenticated session): guard checks current\_user; SET ROLE postgres where the contract requires it.
- Dev-server keep-alive race (ECONNRESET on socket reuse vs 5s idle-FIN): the gate http helper now retries transport errors with backoff (P-8 frozen pattern) - proven root cause: the request never reached the server (log-marked run).
- Teardown robustness: the P9 gate auto-discovers the full FK child surface of a fixture staff row (30 FKs, live-verified) and verifies zero residue; self-heal at start removes leftovers of crashed runs.

## 6. WHERE WE ARE NOW - the live status map

*Snapshot 2026-09-19 (read-only sweep), covering the 2026-09-18 full reflow. FULL REFLOW GREEN - nothing is stopped or waiting: the reflow closed 2026-09-18 15:45, the live invariant sweep passed, and P-9 is FROZEN.*

### 6.1 Gate status (this P-9 reflow round)

| Gate        | Baseline                   | Current (live, 2026-09-18)    | State                                                                       |
|-------------|----------------------------|-------------------------------|-----------------------------------------------------------------------------|
| A           | 39/39                      | 39/39                         | GREEN (re-run post-M7)                                                      |
| E/S         | 54/54                      | 54/54                         | GREEN (re-run post-M7)                                                      |
| F           | 35/35                      | 35/35 (Z1-Z5 visible)         | GREEN after ratified correction (Option A, 2-line swap) - see 6.4           |
| O           | 38/38                      | 38/38                         | GREEN (O-A + O-B ratified ~14:30, applied; residue self-cleared; Section 7) |
| K-L3 / K-L4 | 8/8 / 16/16                | 8/8 / 16/16                   | GREEN (K-L4 with the K4-A 1-line GO)                                        |
| P-1..P-8    | 29/13/25/19/<br>10/15/16/8 | 29/13/25/19/10/15/16/8        | GREEN (P-4: one infra-flake re-run, root-caused, zero file changes)         |
| P-9 (new)   | -                          | 25/25, 0 REAL-RISK, 0 HARNESS | GREEN (final)                                                               |

### 6.2 Live DB invariants (the safety properties that MUST hold)

| Invariant                             | Value now                                                                               | Healthy target           |
|---------------------------------------|-----------------------------------------------------------------------------------------|--------------------------|
| Real staff (DENY53)                   | 53 present, 9 active, 9 pinned                                                          | 53 / 9 / 9               |
| Login pool (active + ACTIVE + pinned) | 9 (live-verified 2026-09-19)                                                            | 9                        |
| staff total                           | 133 (53 real + 80 inert gate fixtures; every non-denylist row fixture-styled per S5.4c) | 53 + inert fixtures only |
| Stale table pointers (prod-wide)      | 0 (live-verified; live pointers total = 0)                                              | 0                        |
| M6 contract FK (RESTRICT)             | 1 live (p9_fk_table_floors_current_order_id)                                            | 1 live                   |
| trg_table_archive_guard               | enabled (tgenabled = O, live-verified)                                                  | enabled                  |
| F residue (tables 101-208)            | 0 (live-verified)                                                                       | 0                        |
| O residue (tables 300-320)            | 0 (live-verified; O-A+O-B re-run self-cleared)                                          | 0                        |
| P9GATE_* fixtures                     | 0 (T.1 verified)                                                                        | 0                        |

*All transient fixture drift (the pool = 13 and the O residue left by the O run that stopped at L211) was self-cleared by the corrected O gate s own START cleanup during the reflow - no manual deletion was ever performed (red line held; no real-staff row was ever touched). The 80 inert staff fixtures are the cumulative F-contract neutralizations of the 14 gate batteries (staff fixtures are never deleted); the P9 gate asserts on every run that none of them is active, pinned, or real-looking outside the 53-ID denylist.*

### 6.3 Git state

```
branch: main      HEAD: the P-9 freeze commit (on top of 6d7a883e, the P-8 freeze)  main == origin
committed in the freeze: M1a-M7 + M1b migrations + rollbacks, .p9-gate.cjs, .p9-audit/, P9_* docs,
  gate corrections (.f-gate.cjs Option A 2-line swap; .o-gate.cjs O-A + O-B; .k-l4-probe.cjs K4-A),
  all 14 reflow reports (.x-gate-report.json), gate run logs, HANDOVER.md P-9 FROZEN block,
  P9_FREEZE_REPORT_2026-09-18.md, this briefing (FROZEN edition)
tree: clean (the red line "no commit before the full reflow is green" was honored)
```

### 6.4 The F conflict - RESOLVED (what happened and why it is closed)

The O run is not the only instance of this class. Gate F (frozen 2026-09-11) crashed its teardown for the same reason: M6 added table\_floors.current\_order\_id to orders ON DELETE RESTRICT, and F s cleanup deleted orders before table\_floors. With ON\_ERROR\_STOP=1 the batch aborted, the crash handler retried into the same FK, and residue was left. The owner ratified Option A (2-line swap: delete table\_floors before orders - pure teardown order, zero new statements, zero new trigger paths; the only DELETE-firing trigger on table\_floors is the archive guard, which the cleanup itself disables). Re-run: 35/35 with Z1-Z5 visible, its own START cleanup cleared its own residue. CLOSED. O is the same class in a second frozen file - hence a second, separate correction cycle (Section 7), per the ratified protocol.

## 6.5 Timeline of this phase (Baku time)

09-17 23:38 P-9 audit ratified; implementation plan written  
09-18 00:00-08:00 M1a-M7 + M1b applied (each: preflight -> ratify -> apply -> verify)  
09-18 00:45-01:10 P9 gate v4 (25 checks) green; F reflow crash (M6 FK vs frozen teardown)  
09-18 01:10 FULL STOP ratified by owner (Option 4) - "contract conflict" framing  
09-18 10:35-11:30 F correction review (source-level evidence); G0: Option A  
09-18 13:00-13:20 F re-run 35/35 GREEN; P9 gate re-run 25/25 GREEN (incl. 2 logged harness calibrations)  
09-18 13:30 O re-run: 36 PASS, STOP at L211 (same FK class); HARD STOP per protocol  
09-18 13:40 O correction review pack delivered; O-A + O-B awaiting G0  
09-18 ~14:30 G0: O-A + O-B (teardown-only); applied  
09-18 14:42-14:44 O 38/38 GREEN (residue self-cleared); K-L3 8/8; K-L4 16/16 (K4-A 1-line G0)  
09-18 15:10-15:39 P-1..P-8 reflow GREEN (29/13/25/19/10/15/16/8; P-4 one infra-flake re-run)  
09-18 15:45 Live invariant sweep: pool = 9, 0 stale/live pointers, 0 F/O/P9 residue  
09-18 15:50 P-9 FREEZE report written; HANDOVER refreshed (freeze commit = step 12)  
09-19 01:30 Briefing (FROZEN edition) regenerated from live re-verification; git commit + push

## 7. The F/O contract conflict - RESOLVED (full record)

*This decision was made 2026-09-18 ~14:30: GO on O-A + O-B. Everything else in the plan then executed unambiguously (Section 8). This section keeps the complete record - the conflict, the F precedent, the fix points, the safety proof, the outcome. Full file-level evidence packs: p9-f-conflict-review/ and p9-o-review/ (delivered in chat); P9\_F\_CORRECTION\_REVIEW\_2026-09-18.md in the repo.*

### 7.1 The conflict, precisely

Two separately-correct, separately-ratified contracts disagree:

- **M6 (P-9, ratified):** table\_floors.current\_order\_id is now a real relational invariant - FOREIGN KEY to orders(id) ON DELETE RESTRICT (orphan = 0 proven before apply; tripwire-gated).
- **Frozen gates F and O (pre-M6):** their fixture teardown deletes the order rows BEFORE the table rows that (by then) point at them. Postgres correctly rejects that delete (RESTRICT), ON\_ERROR\_STOP aborts the batch, the crash handler retries into the same FK, and the gate exits 2 leaving its own fixtures behind.

Classification (ratified by the owner): this is **not** an environment problem and **not** an M6 bug. It is a contract conflict that requires reconciliation. The FK stays (dropping or cascading it was explicitly rejected: it would weaken a ratified safety decision). The frozen gates are corrected through **separate, ratified correction cycles** - teardown order only, zero check-logic changes, zero production surface.

### 7.2 F - RESOLVED (the precedent)

Option A (ratified + applied + proven): inside .f-gate.cjs CLEANUP\_SQL, TWO LINES SWAPPED

```
- DELETE FROM orders WHERE table_number IN (test tables)
+ DELETE FROM table_floors WHERE table_number IN (test tables) <- now first
+ DELETE FROM orders WHERE table_number IN (test tables) <- now second
```

Result: F 35/35 (Z1-Z5 visible), self-cleared its own residue, pool back to 9. git diff = exactly 2 lines.

### 7.3 O - APPLIED (ratified GO): two fix points, both teardown-only

O-A (identical class to F): in .o-gate.cjs CLEANUP\_SQL (L39/L40), swap the two deletes

```
- DELETE FROM orders WHERE table_number IN (300-320 test tables)
+ DELETE FROM table_floors WHERE table_number IN (300-320) <- first
+ DELETE FROM orders WHERE table_number IN (300-320) <- second
```

(O s cleanup already deletes the order CHILDREN first - kitchen\_schedule/outbox/order\_items - i.e. the author already practiced child-first FK discipline; only the M6-era table\_floors pointer was new to the world.)

O-B (one new line above the L211 inline delete):

```
+ S(UPDATE table_floors SET current_order_id=NULL WHERE table_number=312 AND location_id=LOC_A);
S(DELETE FROM orders WHERE order_type='qr_order' AND table_number=312); <- existing, unchanged
```

Why an UPDATE instead of a reorder: the delete is an inline probe cleanup; the table row must survive to the END cleanup/Z1. The pointer must be null BEFORE the order delete - and this exact UPDATE is O s OWN idiom (L198, two lines above: same table, same location filter, same NULL value).

### 7.4 Why O-B is safe (source-level, not "should be fine")

- The pointer-validation trigger fires BEFORE UPDATE OF current\_order\_id only. Its function body, line 1: IF NEW.current\_order\_id IS NULL THEN RETURN NEW; - a NULL pointer passes unconditionally (no orders lookup, no exception path).
- No trigger on table\_floors queries orders on the DELETE path (all 9 trigger functions scanned); the only DELETE-firing trigger is the archive guard, which the cleanup disables two lines above.
- orders has 14 FK children: 7 CASCADE (auto), 6 SET NULL class, 1 RESTRICT (the M6 FK). The other 13 have coexisted with green O runs since before M6 - the M6 FK is the only new blocker.
- The 36 checks that already passed in the stopped O run prove the whole functional surface is intact; only the probe teardown (L211) and END cleanup (L39/40) are M6-unaware.

### 7.5 What the GO did (and what it did NOT touch)

- **Did:** O re-run 38/38 GREEN - its own START cleanup self-cleared the residue (11 tables, 10 orders, 4 OG\_staff neutralized), exactly as predicted; then K-L3 8/8, K-L4 16/16, P-1..P-8 reflow GREEN; then the P-9 FREEZE report + HANDOVER update + git commit/push. The prediction (38/38, self-clear, zero unexpected) was exactly what the run delivered.

- **Did NOT touch:** the M6 FK (stays RESTRICT), any migration, any app code, any trigger, any check/assertion in any gate (O s 38 checks + Z1/Z2 frozen), no manual DB deletion, no git before the full reflow was green.
- **Fallback if O s re-run had surfaced anything unexpected:** the same STOP + evidence + options protocol would have applied; nothing is force-greened. (Not needed - the run was clean.)

## 8. Plan: what happens next (the ratified 12-step sequence)

```
1 [DONE]      Option A applied to .f-gate.cjs (2-line swap, diff = exactly 2 lines)
2 [DONE]      F full-capture re-run
3 [DONE]      F = 35/35 with Z1-Z5 visible (log: .f-gate-run-2026-09-18.log)
4 [DONE]      Post-F read-only assertions (0 residue, pool=9, guard enabled, 0 stale pointers)
5 [DONE]      P9 S5.4 refinement (S5.4a-d) - ratified; + 2 logged harness calibrations (S5.1
               parenthesization; S5.4c regex widened for the G2mu fixture family) - P9-gate-side only
6 [DONE]      P9 gate = 25/25 (0 REAL-RISK, 0 HARNESS; self-heal removed the killed-run fixture)
7 [DONE]      O: G0 on O-A + O-B -> applied -> O = 38/38 (residue self-cleared, as predicted)
8 [DONE]      K-L3 = 8/8
9 [DONE]      K-L4 = 16/16 (with the K4-A 1-line G0)
10 [DONE]     P-1 (29) P-2 (13) P-3 (25) P-4 (19, one infra-flake re-run) P-5 (10) P-6 (15)
               P-7 (16, P7_ALL=1) P-8 (8, P8_ALL=1)
11 [DONE]     P-9 FREEZE report + HANDOVER.md P-9 FROZEN block + this briefing (FROZEN edition)
12 [DONE]     git commit + push (clean tree; main == origin) -> P-9 FROZEN
```

### 8.1 Red lines (unchanged, binding until P-9 freeze)

- No M6 FK change of any kind (no drop, no cascade, no deferrable).
- No manual DB residue deletion - gates clean their own residue via their own START cleanup.
- No migration / app / trigger / view / RLS change outside a ratified cycle.
- No frozen gate check/assertion changes (teardown-only corrections are the sole exception class, each needing its own GO).
- No O/P8 (or any reflow step) before F 35/35 - satisfied; no K/P before O 38/38.
- No commit/push before the full reflow is green.

### 8.2 Risks and mitigations (honest list)

| Risk                                                                      | Likelihood                                                                                                     | Mitigation                                                                                                                                        |
|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Another frozen gate (K/P-series) has a hidden pre-M6 teardown order issue | Low (they did not delete orders with live table pointers in prior green runs; only F and O touch that surface) | Same STOP + evidence + separate correction cycle; each is teardown-scoped                                                                         |
| Dev-server transport flake (ECONNRESET) under gate load                   | Medium (observed; now retried by harnesses that have the pattern)                                              | Frozen retry pattern (P-8) present in P9/O/F harnesses; K-L3 got it in P-5; a harness retry never masks product behavior (root causes are logged) |
| Supabase pooler role anomaly (transient authenticated session)            | Low (observed once)                                                                                            | current_user guard in the P9 gate; migrations use SET ROLE postgres where required; re-runs are safe (idempotent)                                 |
| Fixture drift accumulates (gates neutralize staff, never delete)          | Certain (by F-contract)                                                                                        | P9 gate S5.4b/c assertions: any active/pinned or real-looking name outside the 53-ID denylist is an immediate REAL-RISK; pool must equal 9        |
| Long uncommitted window (git frozen until reflow green)                   | Accepted                                                                                                       | Everything is single-transaction with rollbacks; worst case a future session starts from this briefing + HANDOVER + repo files                    |

## 9. Safety and rollback story (how nothing can be lost silently)

- **Migrations are single-transaction with a rollback file written BEFORE apply.** Every P-9 migration (M1..M7 + corrections) has a matching rollback under supabase/migrations/ (e.g. `_p9_rollback_*`). A failed apply leaves the DB exactly as it was (ON\_ERROR\_STOP + one txn).
- **P-9 FK additions (M6) are pure ADD CONSTRAINT** - each proven orphan = 0 before apply; rollback = DROP CONSTRAINT. No data moved.
- **The M7 staff purge is fully archived:** all 1,055 purged fixture IDs + their child-row sets are recorded in `p9_m7_fixture_ids` (14 archive objects total); `dining_groups` and `merge_tables_v4` were proven intact. The 53 real-staff IDs are byte-verified (DENY53) and re-checked by the P9 gate on every run (S5.4a).
- **Nothing destructive has ever been run without ratification.** The only mass deletion in this project's recent history is M7, which went through a 10-requirement hard-stop review (preflight 24/24, sample rows, orphan proof, archive, rollback, two-way ID verification).
- **Gates are the continuous safety net:** 15 live batteries, each idempotent and zero-residue, each asserting its own invariants (including 0 stale pointers prod-wide, pool = 9, denylist intact). A regression in ANY frozen behavior fails a gate - it cannot be hidden.
- **Current rollback posture for P-9:** if P-9 must be abandoned, the rollback files revert M1..M7 in reverse order (M7 rollback restores the 1,055 fixture rows from archive), and the uncommitted git state can be discarded. The frozen P-1..P-8 surface is untouched by any P-9 rollback. (Not planned - stated for completeness.)

## 10. File map (where everything lives)

| Path (repo root = /Users/mr.apple/saito-admin1)                                                                                                                                                   | What                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| artifacts/saito-admin/                                                                                                                                                                            | The Next.js app: <code>src/app/api/**</code> (180 routes), <code>src/lib/</code> (api-auth, csrf, shiftLock, location-context), <code>.env.local</code> (supabase URL + service_role key - the ONLY credential source) |
| supabase/migrations/                                                                                                                                                                              | All applied migrations (60+); P-9 ones = 2026091800000* + their <code>_rollback_</code> files                                                                                                                          |
| <code>.a-regression.cjs</code> <code>.es-gate.cjs</code> <code>.f-gate.cjs</code> <code>.o-gate.cjs</code><br><code>.k-*-probe.cjs</code> <code>.p1..p8-gate.cjs</code> <code>.p9-gate.cjs</code> | The 15 live gates (frozen except <code>.p9-gate.cjs</code> which is P-9's own harness); each writes a <code>.x-gate-report.json</code>                                                                                 |
| HANDOVER.md                                                                                                                                                                                       | The master handover: frozen baseline, DO-NOT-REOPEN list, residuals, probe conventions (read first)                                                                                                                    |
| ARCHITECTURE.md                                                                                                                                                                                   | SSOT rules, RPC contract, state machines, security model                                                                                                                                                               |
| MASTER_FEATURE_MAP.md /<br>BACKEND_COMPLETION_MAP_2026-09-17.md                                                                                                                                   | A-Z feature inventory / live-and-complete vs partial vs missing modules                                                                                                                                                |
| CANONICAL_SCHEMA_MAP_2026-09-14.md                                                                                                                                                                | Pre-P-7 schema reality (table/function inventory)                                                                                                                                                                      |
| P0..P8 *_2026-09-*.md                                                                                                                                                                             | Per-phase audit + contract + decision-matrix documents                                                                                                                                                                 |
| P9_SCHEMA_AUDIT / P9_IMPLEMENTATION_PLAN /<br>P9_NIGHT_CHECKPOINT /<br>P9_F_CORRECTION_REVIEW / P9_O (repo + chat packs)                                                                          | The current phase: audit, ratified plan, night stop, the F resolution, the O proposal                                                                                                                                  |
| .p9-audit/                                                                                                                                                                                        | P-9 preflight SQL + outputs (raw evidence)                                                                                                                                                                             |
| .db-inventory-2026-09-11.txt / .openapi-live.json                                                                                                                                                 | Live DB and API surface captures (baseline evidence)                                                                                                                                                                   |
| SECRETS.local.md / .credentials                                                                                                                                                                   | Access material - NOT in this PDF, not in git, not in chat logs                                                                                                                                                        |

## 11. Glossary (the terms that recur everywhere)

| Term       | Meaning in this project                                                                               |
|------------|-------------------------------------------------------------------------------------------------------|
| SSOT       | Single Source of Truth - exactly one authoritative table/RPC per business fact                        |
| Atomic RPC | SECURITY DEFINER Postgres function: lock FOR UPDATE, validate, mutate, audit, one txn, JSONB out      |
| Frozen     | Ratified + gated + committed; changes require a separate ratified cycle (gate files AND DB contracts) |

| Term                               | Meaning in this project                                                                                                                                                           |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Gate / probe                       | A re-runnable Node test battery (.cjs) that hits the LIVE DB + dev server, is idempotent, and proves its contract (tallies like 35/35)                                            |
| Reflow                             | Re-running ALL previously frozen gates after any change; identical tallies required, otherwise STOP + triage                                                                      |
| Fixture                            | A test row created by a gate (staff, order, table). F-contract: staff fixtures are neutralized (INACTIVE, pin cleared), never deleted                                             |
| Residue                            | Fixture rows left behind by a crashed gate; must be 0 after a green gate (T.1-style check) and self-cleared by the next START cleanup                                             |
| Pool (login pool)                  | Staff rows that can log in: is_active AND status ACTIVE AND pin set. Healthy value = exactly 9 (the real shift staff)                                                             |
| DENY53                             | The byte-verified list of the 53 real staff IDs (P-9 M7); the P9 gate re-checks it every run                                                                                      |
| FK / RESTRICT / CASCADE / SET NULL | Foreign key + delete behavior. RESTRICT = block parent delete while referenced (the safe default this project chooses); CASCADE = auto-delete children (used only where ratified) |
| Trigger (BEFORE UPDATE OF x)       | A DB function firing on specific column updates; the system's integrity layer (state guards, immutability, pointer validation)                                                    |
| RLS                                | Postgres row-level security; here: service_role = full, authenticated = scoped policies, anon = minimal. RLS-OFF tables are tracked in the completion map                         |
| Pooler anomaly                     | Supabase pooler occasionally maps the session to the authenticated role instead of postgres; guards check current_user and use SET ROLE postgres where the contract requires it   |
| HARNESS-FAILURE vs REAL-RISK       | Gate output classes: test machinery problem (fix harness, retry) vs product invariant violated (fix product via ratified cycle)                                                   |
| Outbox                             | outbox_events rows written inside the business txn; a pump delivers scoped realtime/sync events - DB stays SSOT                                                                   |
| Idempotency key                    | Server-minted (namespace, key) for exactly-once pay/refund; same-key diff-payload = 409 zero-mutation (P-4)                                                                       |
| dirty (table state)                | Post-payment cleanup state, BY DESIGN, off the state registry; new seat blocked until Clear                                                                                       |
| Tripwire                           | A gate check that PROVES a contract is live by attempting a violating write inside a rolled-back txn (e.g. P9 S4.1)                                                               |
| Correction cycle                   | The only legal way to change a frozen gate/contract: file-level proposed diff + source-level evidence + separate GO + re-run + reflow                                             |

## Appendix A - decision log (dated)

| Date (Baku)  | Decision                                                                                                                            | Status                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| 09-17        | P-9 scope boundary ratified (audit + plan); P-9 does not change frozen behavior                                                     | RATIFIED (committed f08798d7)                      |
| 09-18 08:0x  | M7 purge GO (10-requirement review: preflight 24/24, archive, rollback, two-way IDs)                                                | APPLIED + verified (staff 53, pool 9)              |
| 09-18 01:10  | FULL STOP (Option 4) at F crash - contract conflict framing; no frozen-gate edits, no manual deletes, no commits, no M6 FK rollback | RATIFIED                                           |
| 09-18 ~11:00 | F correction: Option A (2-line teardown swap) + P9 S5.4 refinement (S5.4a-d)                                                        | APPLIED + verified (F 35/35, P9 25/25)             |
| 09-18 ~13:30 | O re-run stopped at L211 (same FK class); HARD STOP; O-A + O-B proposed with evidence pack                                          | RATIFIED ~14:30 (GO)                               |
| 09-18 ~14:30 | O correction: O-A (teardown swap) + O-B (NULL-pointer line before the L211 delete) - teardown-only, zero check changes              | APPLIED + verified (O 38/38, residue self-cleared) |
| 09-18 15:50  | Full reflow GREEN (all 14 gates + P-9) -> P-9 FROZEN (freeze report + HANDOVER)                                                     | DONE                                               |
| 09-19 01:30  | P-9 freeze commit + push (step 12); briefing FROZEN edition regenerated from live re-verification                                   | DONE (this document)                               |

## Appendix B - how to run everything (practical)

```
# 0. Prereqs (once): dev server running + psql reachable
cd artifacts/saito-admin && npm run dev # localhost:3000 (must stay up for all gates)
psql -h aws-1-eu-central-1.pooler.supabase.com -p 6543 -U postgres.jbxmlnsicbfkbsatnoej -d postgres
# password: from artifacts/saito-admin/.env.local (PGPASSWORD) - never hardcoded in docs

# 1. P-9 gate (25 checks): cd <repo root> && P9_ALL=1 node .p9-gate.cjs
# 2. The 14 frozen gates (order): node .a-regression.cjs && node .es-gate.cjs && node .f-gate.cjs \
# && O_ROUTE="$(pwd)/artifacts/saito-admin/src/app/api/orders/route.ts" node .o-gate.cjs \
# && node .k-l3-probe.cjs && node .k-l4-probe.cjs && node .p1-gate.cjs && node .p2-gate.cjs \
# && node .p3-gate.cjs && node .p4-gate.cjs && node .p5-gate.cjs && node .p6-gate.cjs \
# && P7_ALL=1 node .p7-gate.cjs && P8_ALL=1 node .p8-gate.cjs
# Expected: 39/54/35/38/8/16/29/13/25/19/10/15/16/8 - any deviation = STOP + triage
# 3. Build + types: cd artifacts/saito-admin && npx next build && npx tsc --noEmit
# 4. Reports land in .x-gate-report.json; run logs are tee-ed to .x-gate-run-*.log by convention
# Conventions: fixtures = P_% / OG_% / FG_% / P9GATE_% (never deleted for staff); test tables =
# high 3-digit ranges; HTTP calls need x-csrf-token == saito_csrf cookie; dev-server ECONNRESET
# under load -> harness retries (frozen pattern).
```

*End of briefing (FROZEN edition). Source documents for deep dives: HANDOVER.md (frozen baseline), the P0-P8 phase documents (per-contract detail), P9\_SCHEMA\_AUDIT + P9\_IMPLEMENTATION\_PLAN + P9\_FREEZE\_REPORT (this phase), and the two correction reviews (the resolved conflict). MASTER\_FEATURE\_MAP.md is the canonical product plan (A-Z master map + Wave roadmap). Ask for the chat-delivered evidence packs for verbatim trigger bodies, FK catalog rows, and gate logs.*